Questions

Q1.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The Big Bang theory is one theory for the origin of the Universe.

The Big Bang theory suggests:

- the Universe had a beginning
- the Universe is still expanding.

Which of these provides evidence that the Universe had a beginning?

(1)

☐ **A** the discovery of other galaxies

☐ **B** the discovery of the moons of Jupiter

☐ **C** the discovery of planets orbiting distant stars

☐ **D** the discovery of cosmic microwave background (CMB) radiation

(Total for question = 1 mark)

Q2.

The Sun is a star that is in a stable state called the main sequence.

Describe the evidence that suggests some stars are moving away from Earth.

(2)

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(Total for question = 2 marks)

Q3.

These are four stages in the evolution of a star similar to the Sun.

They are **not** in the correct order.

1. main sequence star
2. white dwarf
3. red giant
4. nebula

Write down the stages in the correct order.

(2)

The first stage has been done for you.

..... nebula ..... .

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Q4.

Scientists have studied stars to discover how stars evolve.
They know that stars form in a nebula when clouds of dust and gas are pulled together by gravity.
Describe how this process continues for stars much more massive than the Sun.

(4)

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Q5.

A nebula may evolve into a main sequence star, such as the Sun.

Explain how a nebula may evolve into a main sequence star.

(3)

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(Total for question = 3 marks)

Q6.

The Big Bang theory is one theory for the origin of the Universe.

The Big Bang theory suggests:

- the Universe had a beginning
- the Universe is still expanding.

Evidence that the Universe is still expanding comes from observations of light from distant galaxies.

Describe how light from distant galaxies shows that the Universe is still expanding.

(2)

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(Total for question = 2 marks)

Q7.

The wavelength of light emitted from distant galaxies is different when the light is detected on Earth.

Explain how this difference in wavelength shows that the Universe is expanding.

(2)

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(Total for question = 2 marks)

Q8.

Scientists no longer accept the geocentric model of the universe but it was the accepted theory for hundreds of years.

Explain why the evidence available at the time supported the geocentric model.

(3)

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(Total for question = 3 marks)

Q9.

Any object weighs less on the Moon than it does on the Earth.

The gravitational field strength on the Moon is different from the gravitational field strength on the Earth.

Suggest **two** reasons why the gravitational field strength on the Moon is different from the gravitational field strength on the Earth.

(2)

1 .....

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2 .....

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(Total for question = 2 marks)

Q10.

The Hubble Space Telescope is an artificial satellite in orbit about 500 km above the Earth's surface.

This telescope is used to observe the light from very distant objects in the Universe.

Explain why the Hubble Space Telescope must be high above the Earth's surface to make its observations.

(2)

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(Total for question = 2 marks)

Q11.

Which row of the table is correct for both force and velocity?

(1)

	force	velocity
☐ **A**	scalar	scalar
☐ **B**	scalar	vector
☐ **C**	vector	scalar
☐ **D**	vector	vector

(Total for question = 1 mark)

Mark Scheme

Q1.

Question number	Answer	Mark
	The only correct answer is **D** the discovery of cosmic microwave background (CMB) radiation **A** is not correct because it does not indicate the Universe had a beginning **B** is not correct, it is evidence against the geocentric model of the Universe **C** is not correct, it is evidence for other solar systems	**(1)** **AO1**

Q2.

Question number	Answer	Additional guidance	Mark
	An answer that combines the following points of application of knowledge and understanding to provide a logical description: red shift (1) in light from the star (1)	increased wavelength	(2)

Q3.

	Answer	Acceptable answers	Mark
	(nebula) main sequence (star) (1) AND red giant white dwarf (1) All three in correct order for 2 marks	Red Giant White Dwarf (Main sequence) (1)	**(2)**

Q4.

	Answer	Acceptable answers	Mark
	A description including the following stages (up to 3 marks) Protostar (1) Main sequence star (1) (super) red giant (1) supernova (1) neutron star (1) (even more massive star can become) black hole (1) more massive stars have shorter life (1) Three stages in the correct sequence (1)		**(4)**

Q5.

Question number	Answer	Additional guidance	Mark
	An explanation to include any three from: nebula as a cloud of gas/dust (1) gas / atoms pulled together / towards each other (1) by gravitational force (1) temperature increase (1) hot enough for nuclear fusion (1)	 density increase gravity (acting) (resultant) heating s	**(3)**

Q6.

Question number	Answer	Additional guidance	Mark
	A description to include: wavelength (of the light) (1) appears to increase (1) [increase must be linked with wavelength]	 Red shift/Doppler effect (Red shift) shows galaxy moving away accept answers in terms of frequency	**(2)** **AO1**

Q7.

	Answer	Additional guidance	Mark
	(there is) an increase in wavelength (of light) (1) shows galaxies are moving away (from Earth) (1)	allow wavelength stretches allow red shift ignore shift to red end of spectrum ignore objects / stars / planets	**(2)** **AO1**

Q8.

Question number	Answer	Additional guidance	Mark
	An explanation that combines identification – understanding (1 mark) and reasoning/justification – understanding (2 marks): • at the time, there was only naked eye evidence (1) • which indicated Sun/Moon/planets appear to move across the sky (1) • in the same direction, same motion each day (1)	allow valid alternatives, e.g. references to comets	(3)

Q9.

Question number	Answer	Additional guidance	Mark
	Any two of the following 1. mass (1) 2. radius (1) 3. density (1)	allow made of different substance/ material if no other mark awarded, allow 1 mark for (Moon is) smaller or Earth is bigger	(2) AO2

Q10.

Question number	Answer	Additional guidance	Mark
	an explanation linking any **two** from improves clarity (of images) (1) light (from distant objects) is (very) faint (1) idea that atmosphere absorbs/scatters light (1) (HST) is above the atmosphere (1) light pollution (on/from Earth) (1)	accept air for atmosphere	**2** **AO1.1**

Q11.

Question Number	Answer	Mark
	<table><tr><td>**D**</td><td>**vector**</td><td>**vector**</td></tr></table> **The only correct answer is D** ***A*** *'scalar scalar' is incorrect, both force and velocity are vectors* ***B*** *'scalar vector' is incorrect, with force being described incorrectly as a scalar* ***C*** *'vector scalar' is incorrect, with velocity being described incorrectly as a scalar*	**(1)** AO 1 1